INCH-POUND

MIL-C-49285/28
19 October 1989

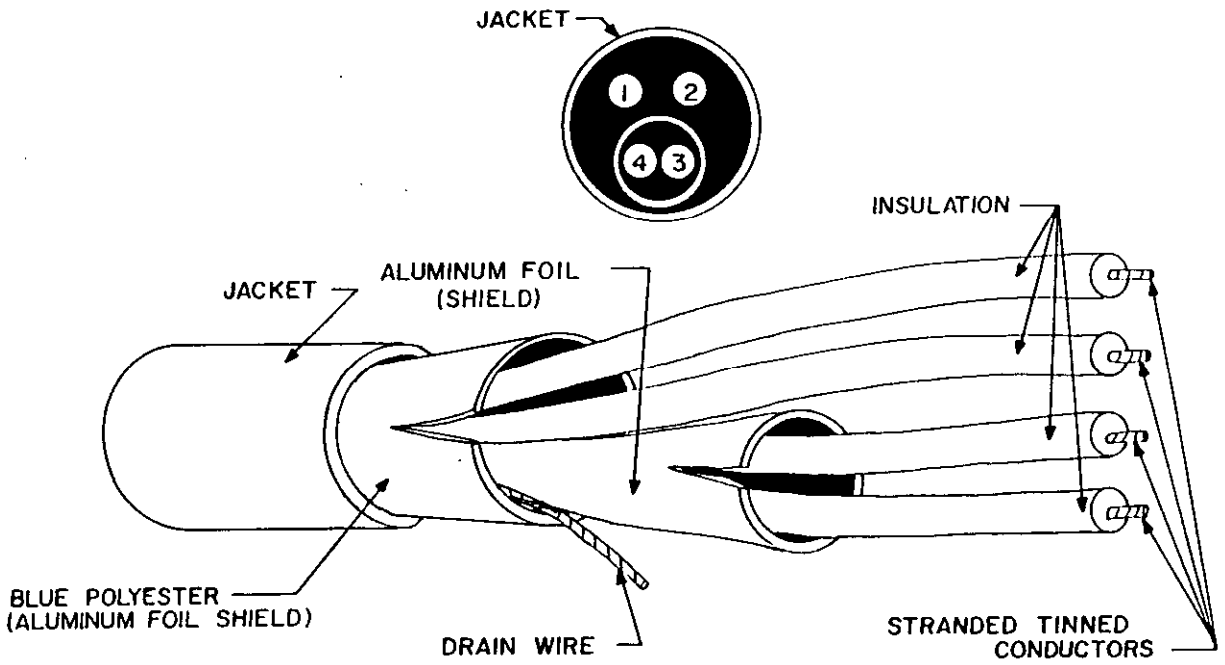
MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, TWO PAIRS OF 25 (7 X 33) AWG, ONE PAIR INDIVIDUALLY SHIELDED

NOTE: NOT FOR AEROSPACE USE.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.



2 PAIR, WITH 1 PAIR INDIVIDUALLY SHIELDED

FIGURE 1. Cable illustration.

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Engineering information: (See table I).

TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	25 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	40 pF/ft
Continuous working voltage	400 V rms maximum between conductors
Shield dc resistance	28 ohms/1,000 feet maximum
Conductor dc resistance	48.0 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of pairs	Two pairs - one pair individually shielded
Conductor type and wire size	Stranded 25 AWG
Conductor stranding	7 X 33 AWG 3 tinned copper 4 tinned coated copper clad steel
Conductor insulation	Polyethylene
Conductor insulation thickness	0.013 inch nominal thickness
Drain wire type and size	Stranded 25 AWG
Drain wire stranding	7 X 33 AWG 3 tinned copper 4 tinned coated copper clad steel
Jacket material	Polyvinyl chloride
Jacket thickness	0.019 inch
Finished cable diameter	0.174 inch maximum outside diameter
Cable style (UL)	Not applicable
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	8.0 twists per foot ±10 percent

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REQUIREMENTS:

Design and construction: (See table II).

Shield location and orientation. This cable consists of a shielded pair and an unshielded pair all enclosed by an overall shield. The polyester aluminum shield associated with the individually shielded pair is to be located on the outer circumference of the pair with the aluminum foil side outward. Insulation on the interior of the shield is to be complete with no aluminum available for contact. The overall shield encompasses the individually shielded pair and the unshielded pair. The overall shield insulation on the exterior of the shield is to be complete with no aluminum available for contact. The overall shield shall have the aluminum foil side inward.

Shield color code. The shield colors shall be blue.

Drain wire location. The drain wire is to be spirally located between the overall shield and the individually shielded pair shield. It is to be in continuous contact with the aluminum surface of both shields throughout the cable. This construction contains one drain wire.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Conductor color code.

1st pair (individually shielded)		2nd pair (under overall shield)	
1st conductor:	Red	1st conductor:	Green
2nd conductor:	Black	2nd conductor:	White

Cabling. Pairs cabled on a common axis to reduce diameter.

Shield effectiveness evaluation test connections. The cable will be tested as if it were one pair. The test shall be performed with the jacket in place. Install in the test fixture. Test one pair with the other pair floating. Alternate the connections in order to test the second pair. Care must be taken in order to preserve both shields during testing.

Shield integrity test.* The following list of responses refer to the pair located under the overall shield. This pair does not have an individiual pair shield covering it.

Frequency	Response **
100 kHz	115 dB below reference
500 kHz	105 dB below reference
1 MHz	100 dB below reference
5 MHz	84 dB below reference
10 MHz	76 dB below reference
15 MHz	71 dB below reference
20 MHz	67 dB below reference
25 MHz	63 dB below reference
30 MHz	60 dB below reference

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 45 dB below reference.

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The following list of responses refers to the pair covered by an individual pair shield and located within the overall shield.

Frequency	Response **
100 kHz	115 dB below reference
500 kHz	105 dB below reference
1 MHz	101 dB below reference
5 MHz	89 dB below reference
10 MHz	85 dB below reference
15 MHz	81 dB below reference
20 MHz	76 dB below reference
25 MHz	71 dB below reference
30 MHz	66 dB below reference

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 47 dB below reference.

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Durometer hardness. The cable shall have a durometer hardness of 85 ±5.

Part of Identifying Number (PIN): The PIN shall be M49285/28.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Air Force - 71
DLA - ES, IS

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1114-29)

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